Abstract

This project presents a comprehensive and systematic approach to text-based emotion classification across six categories: anger, fear, joy, love, sadness, and surprise.

The study details a complete data science pipeline, beginning with initial data exploration, robust text preprocessing, and addressing significant class imbalance using advanced sampling techniques such as SMOTE and ADASYN.

A comparative analysis was performed, pitting traditional machine learning models (TF-IDF with Logistic Regression/Naive Bayes) against modern deep learning architectures.

The optimal performance was achieved with an Advanced Neural Network employing a Bidirectional LSTM (BiLSTM) layer and a custom Attention Mechanism, which successfully achieved 93% accuracy. The project demonstrates the superiority of attention-based deep learning models in capturing nuanced emotional context in short text, providing a highly effective and interpretable solution for multi-class emotion recognition.

Keywords: Emotion Classification, Natural Language Processing (NLP), Deep Learning, BiLSTM, Attention Mechanism, Class Imbalance, Text Classification, Sentiment Analysis

CHAPTER 4 PROJECT WORK

4.1 Background and Context

To build an efficient multi-class emotion classification system that identifies emotional tone in text using traditional ML/ deep learning models.

Classifying text into 6 distinct emotion categories: anger, fear, joy, love, sadness, and surprise

Dataset contains ~20,000 samples with significant class imbalance (surprise: 3.58%, love: 8.15% vs joy: 33.51%, sadness: 29.16%)

4.2 Problem Statement

Short text expressions lack sufficient contextual information compared to longer documents
Traditional bag-of-words approaches fail to capture sequential patterns and emotional nuances

Key Problems Identified:

- Class Imbalance: Minority classes (surprise, love) severely underrepresented, leading to poor model performance on these categories
- Loss of Context: Traditional ML methods ignore word order and sequential dependencies crucial for emotion detection
- Semantic Understanding: Need to capture not just individual words but their relationships and contextual meaning
- Negation & Intensity: Phrases like "not happy" or "very sad" require understanding beyond individual tokens

4.3 Objectives

Primary Objective:

Develop an optimal emotion classification system that achieves balanced performance across all six emotion categories.

Specific Objectives:

1. Establish Baseline Performance
   - Implement traditional ML approaches (TF-IDF + Logistic Regression/Naive Bayes)
   - Identify baseline accuracy and per-class performance metrics
2. Address Class Imbalance
   - Apply oversampling techniques (SMOTE, ADASYN) to balance training data
   - Improve minority class recall without sacrificing overall accuracy
3. Optimize Traditional ML Performance
   - Conduct systematic hyperparameter tuning using GridSearchCV
   - Explore n-gram features (unigrams, bigrams) for phrase-level patterns
4. Transition to Deep Learning
   - Implement sequential models (BiLSTM) to capture word order and context
   - Leverage pre-trained embeddings (Word2Vec, GloVe) for semantic understanding
5. Enhance Model Interpretability
   - Implement attention mechanisms to identify emotionally relevant words
   - Provide explainable predictions showing which words contribute to emotion classification
6. Achieve Optimal Performance
   - Systematically compare all approaches (10+ model variations)
   - Achieve high accuracy with balanced F1-scores across all emotion classes
   - Target specific improvement for minority classes (surprise, love)

4.4 Project Flow Chart

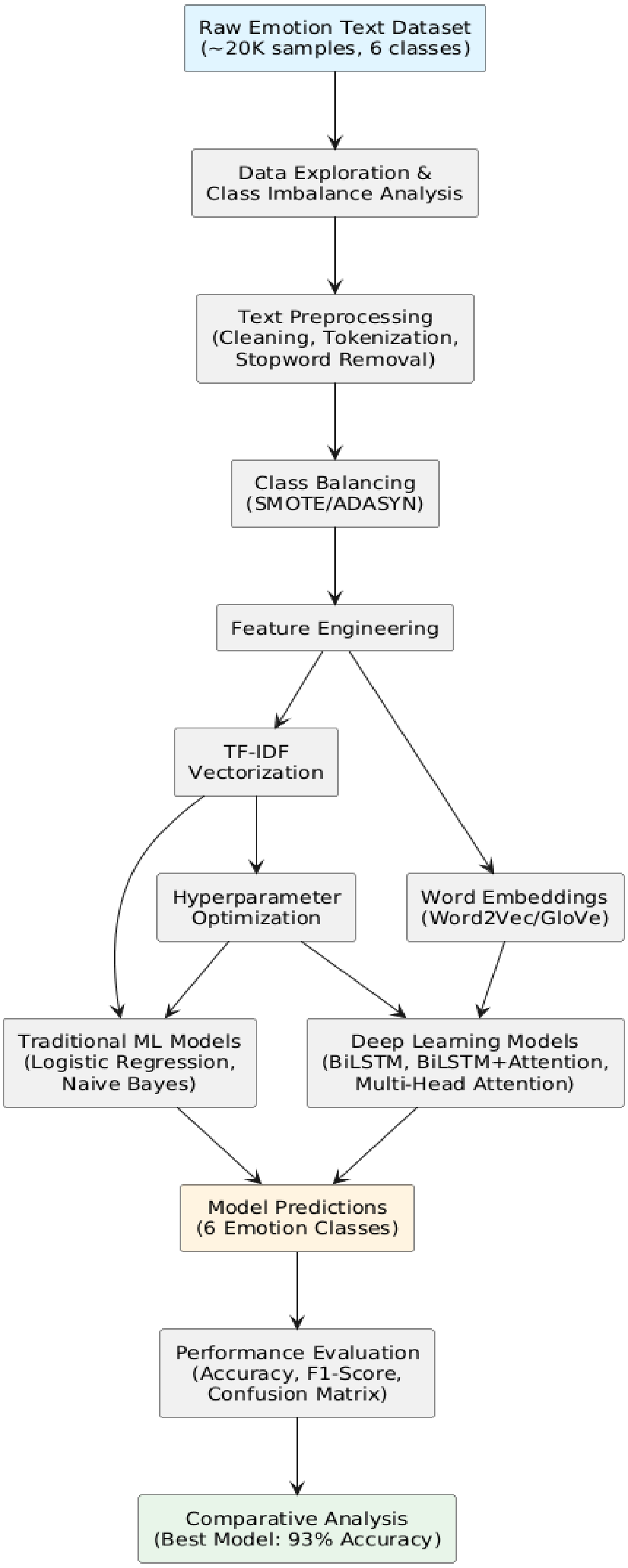

Fig. 4.4.1 Project Flow

4.5 Methodology Overview

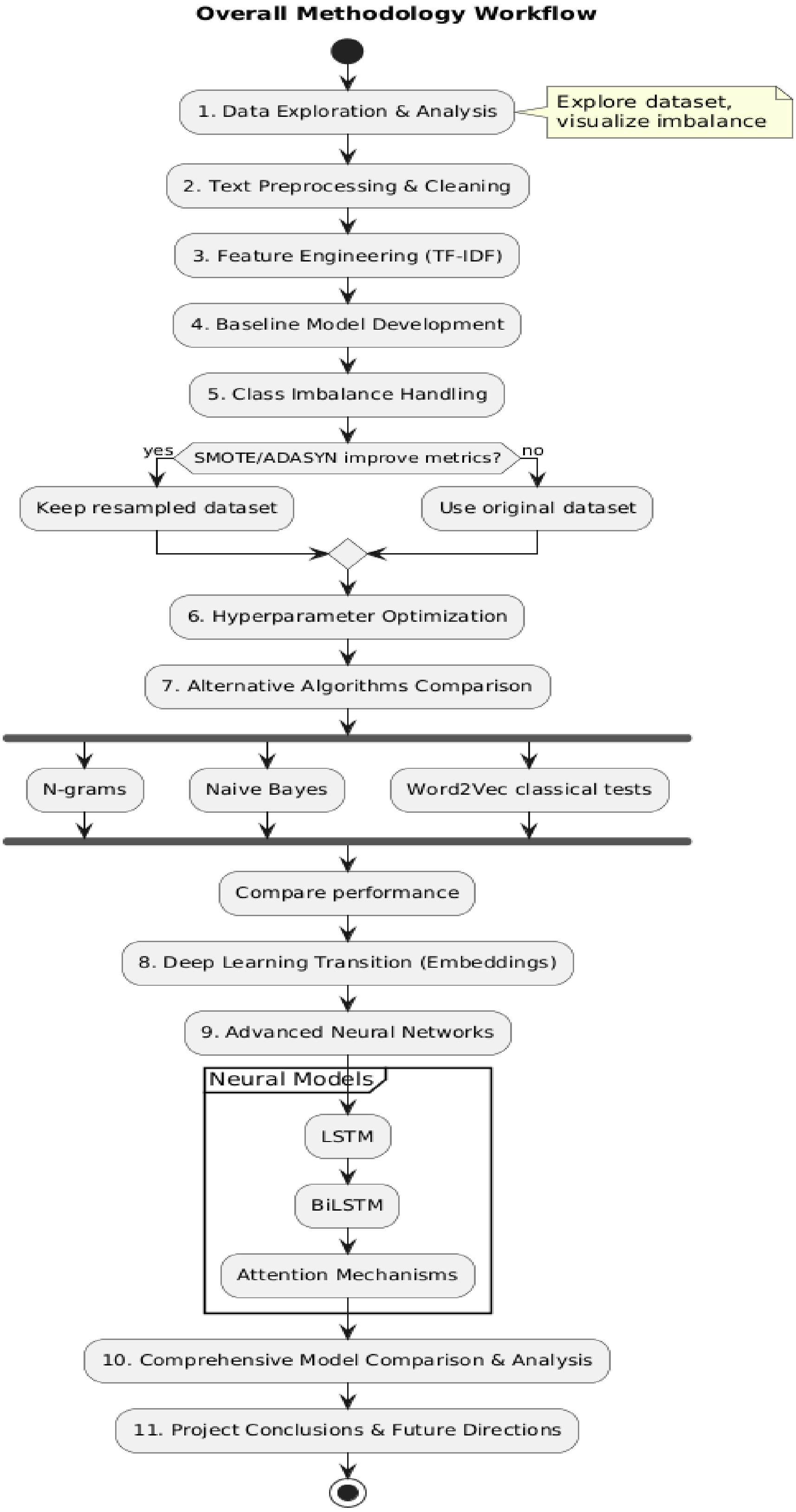

Fig. 4.5.1 Methodology

4.6 Implementation & Testing

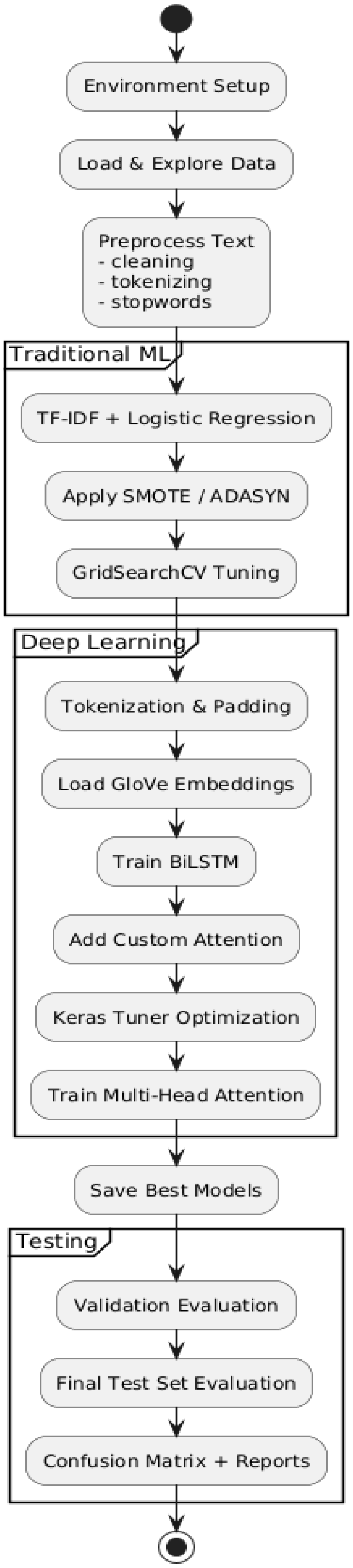

Fig.4.6.1 Implementation Steps

4.7 Dataset Characteristics

- Classes: 6 emotion categories (anger, fear, joy, love, sadness, surprise)
- Challenge: Significant class imbalance with 'surprise' and 'love' being minority classes
- Data Split: Training (~16K), validation (~2K), and test (~2K) sets
- Text Type: Short emotional expressions and statements

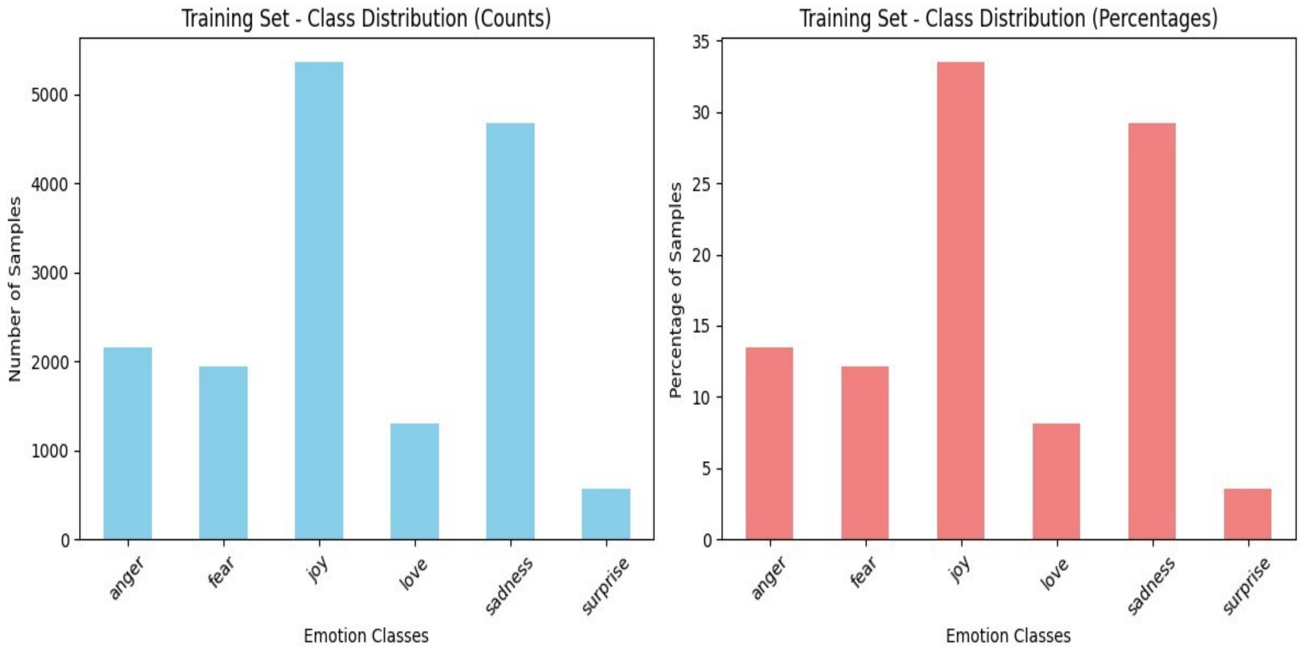

Fig. 4.7.1 Class Distribution

4.8 Feature Analysis

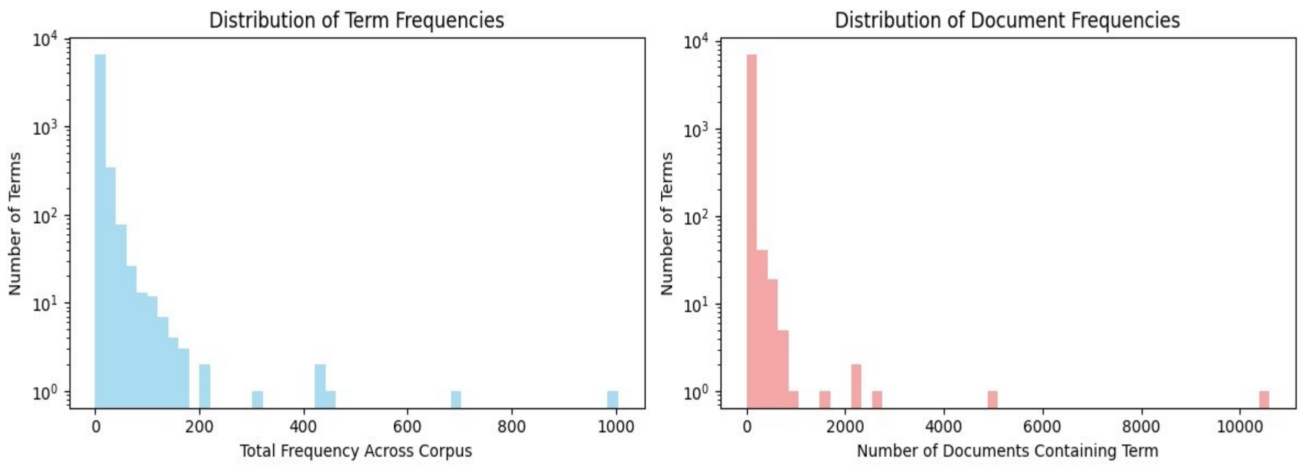

Fig. 4.8.1 Term Frequency Distribution

CHAPTER 5 RESULTS AND DISCUSSION

5.1 Classical Machine Learning Results

TF-IDF + Logistic Regression provided high accuracy for dominant emotion classes but limited performance for minority classes due to feature sparsity and imbalance.

5.2 Performance Analysis and Insights

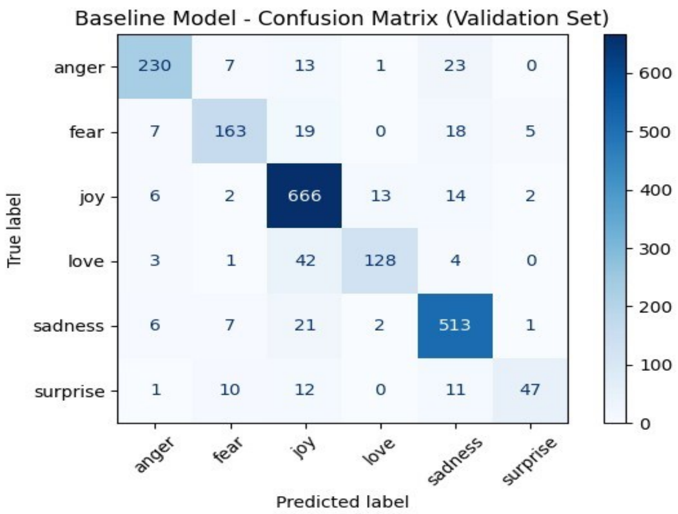

Fig. 5.2.1 Confusion Matrix (Baseline Model)

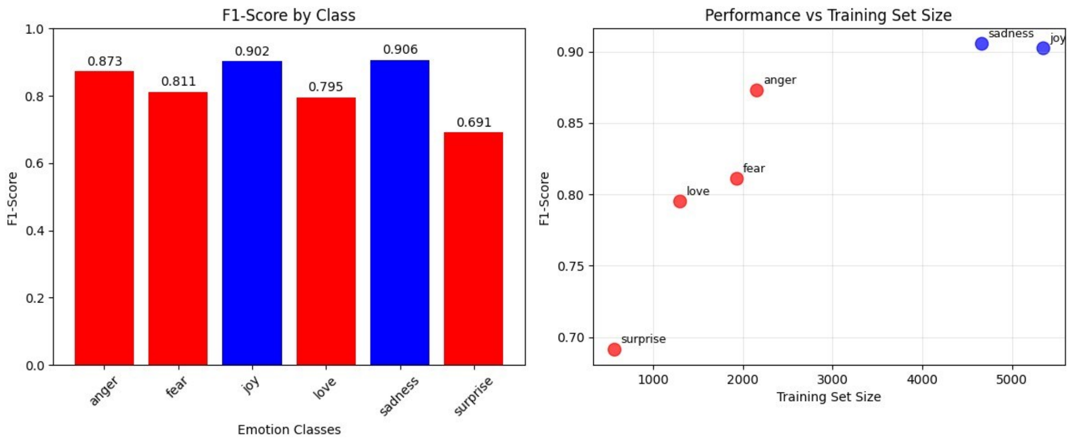

Fig. 5.2.2 Baseline Model Performance

Key Observations:

- Red points/bars indicate minority classes identified earlier
- Clear correlation between class size and model performance
- Minority classes (love, surprise) show significantly lower F1-scores
- This indicates a need for class imbalance handling techniques

5.3 Deep Learning Results

BiLSTM with attention improved contextual modeling and delivered better recall in minority classes through sequential pattern extraction.

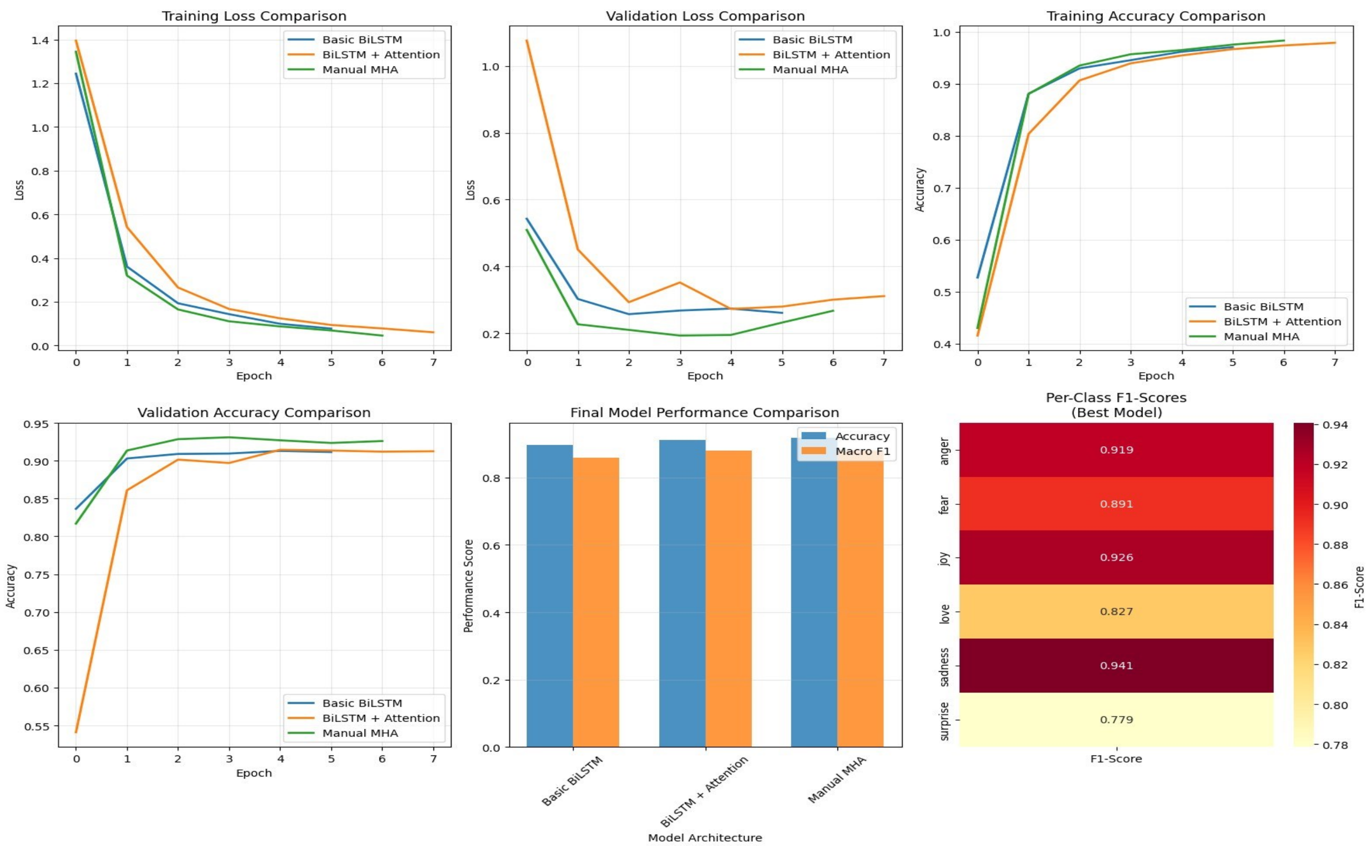

Fig. 5.3.1 DL Model Comparison

5.4 Advanced Architectures Explored

A. BiLSTM (Bidirectional LSTM)

a. Processes text in both forward and backward directions
b. Captures long-range dependencies and context
c. Better understanding of sentence structure

B. BiLSTM + Attention

a. Adds attention mechanism to focus on relevant words
b. Improves interpretability and performance
c. Handles variable-length sequences effectively

C. Multi-Head Attention (MHA)

a. Multiple attention heads capture different aspects
b. Parallel processing of various linguistic patterns
c. State-of-the-art performance for sequence modeling

CHAPTER 6 CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The project successfully demonstrated the evolution from traditional ML methods to deep-learning based architectures for emotion classification.

A complete pipeline for emotion classification was implemented, moving from traditional machine learning to advanced deep learning models.

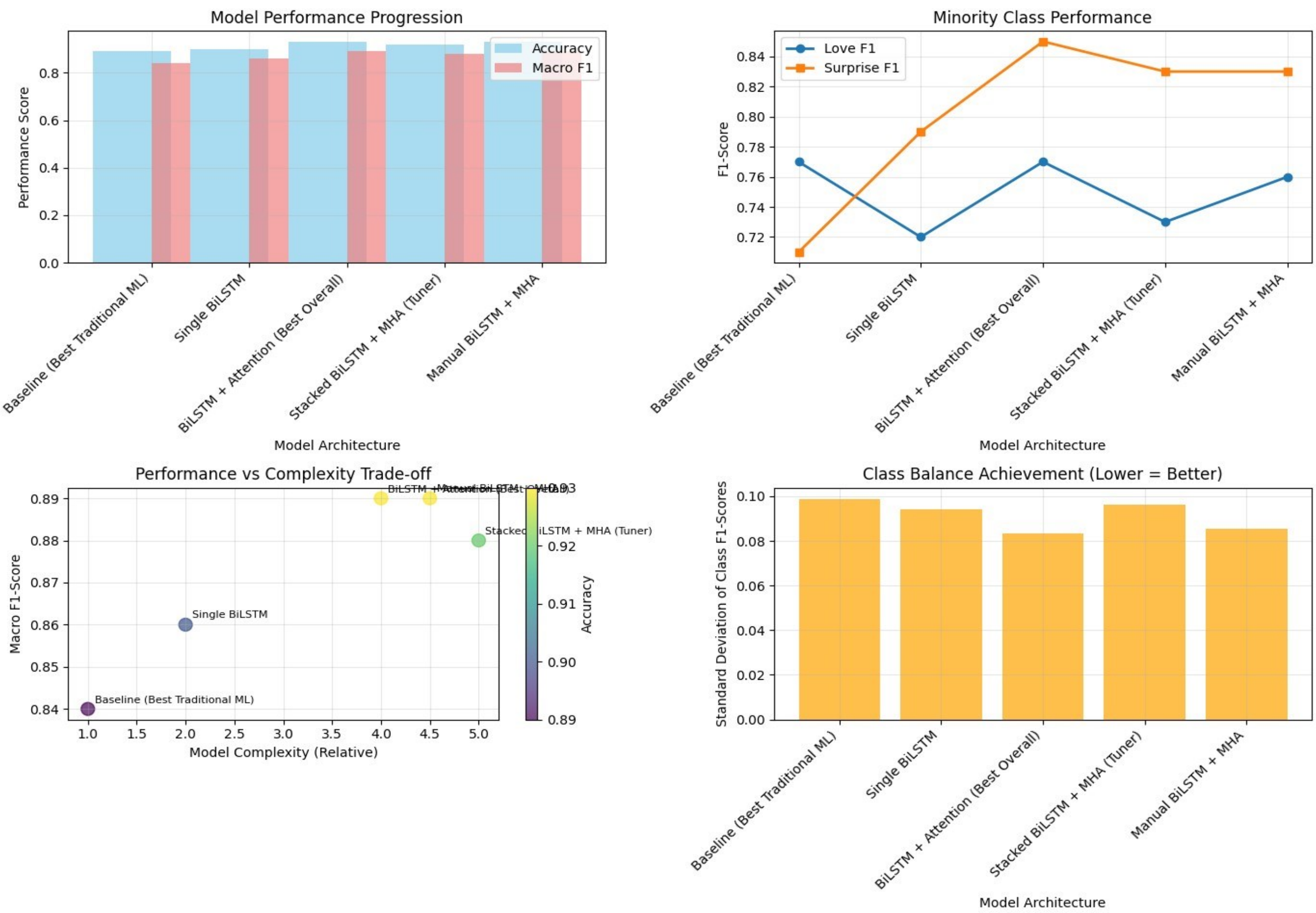

Fig. 6.1.1 Summary Analysis

The results showed that while TF-IDF and Logistic Regression provide a strong baseline, they struggle with class imbalance and subtle emotional cues.

Oversampling techniques like SMOTE and ADASYN improved minority-class recall, but the most significant gains came from neural models.

The BiLSTM with attention emerged as the best-performing approach, achieving high accuracy and balanced performance across classes.

Traditional ML Limitation	Deep Learning Solution
Ignores word order (bag-of-words)	BiLSTM captures sequential context
Simple averaging weakens nuance	Attention mechanisms focus on important words
Poor generalization on minority classes	Class weights + contextual understanding
Static word representations	Dynamic context-aware embeddings
Limited feature interactions	Neural networks learn complex patterns

Table 6.1.1 ML vs DL in NLP

Overall, the project demonstrates that contextual sequence models are far more effective for capturing emotional nuance compared to classical ML methods.

6.2 Summary Table

Model	Accuracy	Macro F1	Surprise F1	Love F1	Sadness F1
Word2Vec + ADASYN + LogReg	0.41	0.24	0.17	0.16	0.45
GloVe + ADASYN + LogReg	0.57	0.47	N/A	N/A	N/A
Single BiLSTM	0.90	0.86	0.72	0.79	0.94
BiLSTM + Attention (Current Best)	0.93	0.89	0.77	0.85	0.97
Stacked BiLSTM + MHA (Tuner)	0.92	0.88	0.73	0.83	0.96
Manual BiLSTM + MHA	0.93	0.89	0.76	0.83	0.96

Table 6.2.1 Summary of all models

6.3 Future Scope

- Integrate multimodal emotion understanding (text + audio + video)
- Future work can explore transformer-based models such as BERT or RoBERTa for deeper contextual understanding and improved minority-class performance.
- Expanding the dataset or applying targeted data augmentation may further reduce imbalance issues.

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